

AI-Generated vs. Authentic Etsy Image Detection: Literature Review and Project Report

Avinash Rajasekar
A00051626
School of Computing
Dublin City University
Dublin, Ireland

Saran Nagarajan
A00037220
School of Computing
Dublin City University
Dublin, Ireland

Abstract—We developed a binary image classifier to identify AI-generated versus authentic product images on Etsy listings. Starting from a simple raw-pixel Random Forest baseline ($F1 \approx 0.62$), we iteratively explored feature-based and deep learning models. Key stages included extracting features from the pre-trained MobileNetV2 CNN and evaluating multiple classifiers, with an SVM achieving the best result ($F1 \approx 0.76$), followed by frozen DINOv2 vision-transformer features with an SVM ($F1 \approx 0.86$), and progressive end-to-end fine-tuning of DINOv2 across two, four, and eight transformer blocks ($F1 \approx 0.91, 0.931$, and 0.9409 respectively). We applied standard data augmentation and validation-based threshold tuning to maximize F1, achieving a final score of 0.9409 . Our findings align with prior work showing that large pretrained backbones and fine-tuning can substantially improve synthetic-image detection performance [1]. However, detection remains challenging as generators improve and shift beyond the training distribution [7], [11]. We conclude with recommendations including cross-validation, spectral feature fusion, continual dataset updates, and diffusion-aware detection.

Index Terms—AI-generated image detection, image forensics, Etsy, DINOv2, MobileNetV2, transfer learning, computer vision

I. INTRODUCTION

The recent rise of generative AI has made it easy for users to create highly realistic product images using systems built on GANs, diffusion models, and related architectures. In e-commerce settings such as Etsy, this creates a practical trust problem: buyers often expect authentic photos of real products, yet synthetic images may now be visually convincing enough to pass as genuine. This motivates the need for forensic detection systems that can distinguish AI-generated images from real photographs.

This problem falls within the broader field of synthetic-image forensics, which aims to identify whether an image has been generated or manipulated by an algorithm rather than captured by a camera [1], [5], [10]. The task is becoming harder as generation quality improves and as newer systems reduce the obvious visual artifacts that earlier detectors relied upon [7], [11]. In this project, we focus specifically on *fully generated* product images rather than local image edits or face swaps. This distinction matters because the forensic cues for fully synthetic images differ from those used in traditional deepfake or manipulation detection.

II. PROBLEM BACKGROUND

A. Generative Models

Generative Adversarial Networks (GANs) introduced adversarial training as a powerful mechanism for realistic image synthesis [3]. Diffusion models later advanced the field further by iteratively denoising random noise into structured images [4]. Commercial systems such as DALL-E, MidJourney, and Stable Diffusion build upon these broader developments in generative modeling, and their outputs have become sufficiently realistic to create challenges for authenticity verification [7].

Unlike deepfake systems that alter parts of existing photos, the synthetic images considered in this study are fully generated. Every pixel is synthesized by the model rather than recorded from a real object or scene. As a result, the classification problem is not simply a matter of detecting manipulated regions, but of learning the global statistical differences between real and synthetic imagery.

B. Detection Challenges

Synthetic-image detectors typically rely on subtle artifacts in the spatial domain, frequency domain, or both [5], [10], [11]. Nataraj *et al.* showed that color co-occurrence statistics can reveal GAN artifacts when coupled with a CNN classifier [5]. Wang *et al.* later demonstrated that a ResNet-50 detector trained on one generator family could generalize surprisingly well to others by exploiting common statistical traces [10]. However, this generalization is not guaranteed. Ren *et al.* reported that many open-source detectors perform poorly on newer, stronger generators unless they are adapted or retrained [7]. Similarly, Wang *et al.* proposed DIRE specifically because conventional detectors struggled with diffusion-generated images [11]. These findings underscore the need for robust representations, broader training data, and continual detector updates.

III. DATASET AND PREPROCESSING

The dataset provided for this project consisted of `train.csv` containing `image_id` and binary labels, `test.csv` containing only image IDs, and a ZIP archive containing the corresponding images. File paths were built from the image IDs, and no missing images were found after

mapping. The code output from `train_df.shape` and class counts suggested a moderately large and roughly balanced binary dataset.

Images were resized and normalized to match model requirements. Initial experiments used low-resolution inputs of 32×32 to enable a fast baseline. Later experiments used 224×224 inputs to match the expected size of common pretrained models such as ResNet, MobileNetV2, EfficientNet, and DINOv2. We also explored Fourier-based statistics, motivated by prior work showing that frequency-domain cues can help detect synthetic imagery. In practice, deep learned features outperformed the simple FFT statistics we tested [5] [11].

Standard augmentations were applied during training, including random horizontal flips, small rotations, random crops, and color jitter. These transformations were used to improve robustness and reduce overfitting. Standard augmentations were applied during training, including random horizontal flips, small rotations, random crops, and color jitter. These transformations were used to improve robustness and reduce overfitting. Test-time augmentation (TTA) was explored during experimentation, but the final best-performing model did not rely on TTA.

IV. MODEL ARCHITECTURES

Table I summarizes the main model families considered in this project.

Each model was used in one of two ways:

- 1) **Feature extractor:** the pretrained backbone was frozen and used to generate embeddings, with an SVM trained on top.
- 2) **Fine-tuning:** some or all backbone layers were unfrozen and trained end-to-end on the target task.

The SVM setup is supported by the classical theory of support-vector networks, which shows that high-dimensional feature mappings can provide strong generalization when paired with a margin-based classifier [2].

V. FEATURE EXTRACTION VS. FINE-TUNING

Pretrained features provided a strong starting point. Using MobileNetV2 as a frozen feature extractor, we evaluated multiple classifiers on the extracted embeddings, including Logistic Regression, Random Forest, XGBoost, and an SVM. The SVM with an RBF kernel yielded the best result at approximately $F1 = 0.76$, demonstrating that even lightweight pretrained CNN features can capture useful forensic cues [8]. Logistic Regression and Random Forest on the same features produced lower scores, and XGBoost offered no meaningful improvement over the SVM.

We then moved to transformer-based features using DINOv2, a self-supervised vision transformer trained on massive image collections to learn broadly transferable visual representations. Using frozen DINOv2 features with an SVM raised performance to approximately $F1 = 0.86$, indicating that feature quality was a major performance bottleneck rather than the choice of classifier [6].

The largest gains came from end-to-end fine-tuning. We progressively unfroze DINOv2 transformer blocks, beginning with the final two blocks at a learning rate of 1×10^{-4} , which yielded approximately $F1 = 0.91$. Extending to four unfrozen blocks with a reduced learning rate of 5×10^{-5} improved this to approximately $F1 = 0.931$. Finally, unfreezing eight blocks at 3×10^{-5} with a linear warmup followed by cosine decay scheduling produced the best validation score of $F1 = 0.9409$. These results align with prior evidence that stronger backbones and task-specific fine-tuning can substantially improve AI-image detection [1], [10].

VI. TRAINING STRATEGY

We used an 80/20 train-validation split stratified by label. Early stopping was applied to prevent overfitting, although full K -fold cross-validation was not implemented due to time constraints. For PyTorch-based fine-tuning of DINOv2, we used AdamW. For smaller Keras-based experiments, we used Adam. Initial learning rates were around 10^{-4} for top layers and were lowered to approximately 5×10^{-5} when more layers were unfrozen and lowered to $3e-5$ (8 blocks).

Batch size was fixed at 32 throughout all experiments. For MobileNet-based experiments, training was limited to roughly five epochs with early stopping because the dataset was not extremely large and convergence was relatively fast. For DINOv2, training was carried out in phases: an initial phase with two blocks unfrozen, followed by a lower-learning-rate phase with four blocks unfrozen. Model checkpoints were selected based on the best validation $F1$ score.

Binary cross-entropy was used as the training loss. The primary evaluation metric was $F1$ score, as required by the project. Since threshold selection can materially affect $F1$, we tuned the decision threshold on the validation set rather than fixing it at 0.5.

VII. DATA AUGMENTATION

During training, we applied standard data augmentation including random horizontal flips, small rotations (approximately $\pm 15^\circ$), random crops, and color jitter.

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Test-time augmentation (TTA) was also explored as an additional strategy, where predictions were averaged across multiple transformed versions of each image. While this provided minor improvements in some experiments, it was not used in the final best-performing model.

VIII. EVALUATION METRIC AND THRESHOLD SELECTION

We used $F1$ score as the main evaluation metric because it balances precision and recall. This is suitable for a binary detection task where false positives and false negatives are both meaningful. We also tracked precision and recall separately in order to understand trade-offs more clearly.

TABLE I
COMPARISON OF CANDIDATE MODEL ARCHITECTURES

Model	Parameters	Input Size	Pros	Cons
MobileNetV2 [8]	3.4M	224×224	Lightweight, fast, and suitable for feature extraction	Lower capacity than larger CNNs and transformers
EfficientNetB0 [9]	~ 5.3 M	224×224	Strong accuracy-to-parameter trade-off; effective in prior AI-image detection studies [1]	Heavier and more complex than MobileNetV2
DINOv2 ViT [6]	Base: ~ 86 M; Large: ~ 307 M	224×224	Powerful self-supervised features; strongest overall performance in our experiments	Very large model; slower training and higher GPU memory usage

TABLE II
VALIDATION PERFORMANCE ACROSS METHODS

Method	Validation F1
Raw Pixels + Random Forest	0.62
MobileNetV2 Features + SVM	0.76
DINOv2 Features + SVM	0.86
DINOv2 Fine-Tuned (2 Blocks)	0.91
DINOv2 Fine-Tuned + TTA	0.92
DINOv2 Fine-Tuned (4 Blocks)	0.931
DINOv2 Fine-Tuned (8 Blocks + warmup scheduling)	0.9409

IX. EXPERIMENTAL RESULTS

Table II presents a summary of the main validation results.

These results confirm that feature quality was the main limiting factor. Upgrading from MobileNetV2 to DINOv2 produced a large increase in performance, and fine-tuning the transformer improved results further. The final DINOv2 model outperformed the raw-pixel baseline by more than 0.30 F1 and outperformed the MobileNetV2 feature pipeline by about 0.17 F1.

The results are consistent with prior work. Baraheem and Nguyen found that strong pretrained backbones such as EfficientNet perform well for AI-image detection [1]. Wang *et al.* showed that synthetic-image artifacts can generalize across model families [10]. Our results extend that general pattern by showing that a modern self-supervised transformer backbone can be even more effective than a lightweight CNN feature extractor for this task.

X. QUALITATIVE ANALYSIS

We inspected misclassified examples to identify common error patterns. Most false positives were highly polished or artistic real images that the model appeared to interpret as “too perfect” or overly stylized. Most false negatives were low-quality, blurred, or visually noisy AI-generated images, where obvious synthetic cues were less visible.

This error profile suggests that the classifier relied on distributed visual statistics rather than single obvious markers. Such behavior is broadly consistent with explainable AI findings from synthetic-image detection work, where activation maps often highlight subtle textures, backgrounds, and local inconsistencies rather than only the main subject [1].

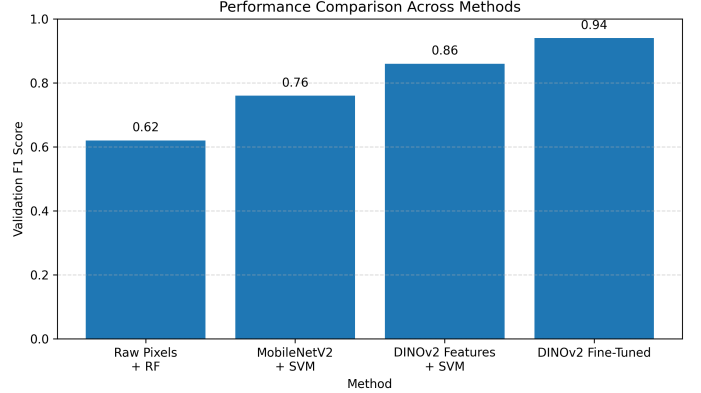


Fig. 1. Figure X visualizes the performance improvements across methods, highlighting the substantial gains from DINOv2-based approaches.



Fig. 2. Figure X shows representative examples of correct and incorrect predictions, where T denotes the ground truth and P denotes the model prediction

XI. LIMITATIONS AND CRITIQUE

Although the final model achieved a strong validation F1 score, several limitations remain.

A. Generalizability

The model was trained on a fixed dataset that may not capture future generation styles. Ren *et al.* showed that detector performance can drop sharply on images from unseen and more advanced generators [7]. This means periodic retraining or continual dataset expansion is likely necessary.

B. Model Complexity

Our best model, DINOv2, is large and computationally expensive. The base model contains roughly 86 million parameters and requires substantial GPU memory for training and inference [6]. In contrast, MobileNetV2 is much cheaper to run but delivered weaker performance [8]. This trade-off

suggests that compression or distillation would be useful for deployment.

C. Adversarial Robustness

We tested only mild perturbations such as blur and compression. More targeted adversarial attacks may significantly degrade performance. Recent work on diffusion-aware detection, especially DIRE, suggests that reconstruction-based signals may provide greater robustness to some generator families and post-processing pipelines [11].

D. Evaluation Bias

Threshold tuning was performed on the validation set, which assumes that the validation class ratio matches deployment conditions. If the real-world prevalence of AI-generated listings shifts, the optimal decision threshold may also shift. In addition, F1 score implicitly balances precision and recall equally, which may not match the needs of a production moderation system.

XII. COMPARISON WITH LITERATURE

Our findings are broadly consistent with recent literature on synthetic-image detection. CNN-based methods remain effective because they capture local texture and artifact cues [1], [5], [10]. In our experiments, MobileNetV2 features provided a strong intermediate baseline, reflecting the usefulness of pretrained convolutional features [8]. The success of EfficientNet-based transfer learning in prior work also supports the broader conclusion that stronger pretrained backbones improve detection performance [1], [9].

A key difference in our work is the use of DINOv2, a modern self-supervised transformer, rather than only conventional supervised CNNs [6]. This choice aligns with the growing trend toward large pretrained representation learners for downstream visual tasks. Our results suggest that transformer representations can be particularly effective for detecting subtle synthetic artifacts in product images.

We did not implement explicit frequency-domain detectors or watermark-based methods. However, prior work indicates that spectral cues and reconstruction signals can be complementary to spatial features [5], [11]. Future systems could benefit from combining these different sources of evidence in a single ensemble.

XIII. RECOMMENDATIONS AND NEXT STEPS

Based on both the project findings and the broader literature, the following next steps are recommended:

- **Cross-Validation and Ensembling:** Use K -fold cross-validation to estimate variance more reliably and ensemble multiple strong models for improved robustness.
- **Feature Fusion:** Combine spatial deep features with spectral or co-occurrence features to capture complementary forensic signals [5], [11].
- **Expand Training Data:** Continuously update the dataset with images from newly released generators.

TABLE III
RECOMMENDED NEXT STEPS

Next Step	Description	Effort
Cross-Validation & Ensemble	K -fold CV and model ensembling	Medium
Feature Fusion	Add FFT or co-occurrence features	Medium
Expand Training Data	Add samples from new AI generators	High
Diffusion-Based Detector	Use DIRE-style reconstruction error [11]	High
Model Compression	Distillation or pruning	Medium
Adversarial Training	Train on perturbed inputs	Medium
Explainability Tools	Saliency maps or SHAP	Low

- **Diffusion-Based Detection:** Implement a diffusion-aware detector such as DIRE to better handle diffusion-generated imagery [11].
- **Model Compression:** Distill the transformer into a smaller network or apply pruning for faster inference.
- **Adversarial Training:** Train on perturbed or post-processed inputs to improve real-world robustness.
- **Explainability Tools:** Add saliency methods or SHAP-style explanations to support manual review and error analysis.

Table III summarizes these recommendations.

XIV. CONCLUSION

This project demonstrates that AI-generated versus authentic Etsy image detection can be effectively addressed using modern pretrained visual models. While a raw-pixel Random Forest baseline provided limited performance, pretrained CNN features significantly improved results, and DINOv2-based transformer representations delivered further gains. The strongest model, obtained through fine-tuning DINOv2, achieved a validation F1 score of 0.9409.

Overall, the results support existing literature indicating that backbone quality, pretrained representations, and task-specific fine-tuning are critical for strong performance in synthetic-image detection [1], [6], [10]. At the same time, challenges remain in generalization to unseen generators, robustness to post-processing, and deployment efficiency.

Future work should focus on expanding training datasets, integrating hybrid spatial-frequency features, and developing models capable of continual adaptation to evolving generative techniques. [7], [11].

XV. CONTRIBUTIONS

- **Avinash Rajasekar:** Contributed to the research, advanced experimentation, and final documentation of the project. His work included reviewing related research papers, preparing the literature review, and contributing to the abstract, introduction, methodology discussion, and conclusion. He was also responsible for the DINOv2-based experiments, including feature extraction with shallow

classifiers, fine-tuning with different unfreezing strategies, threshold tuning, validation-based performance evaluation, and generation of the final submission.csv file. He further contributed to interpretation of results and final proofreading of the report.

- Saran Nagarajan: Contributed to the implementation and experimental development of the project. His work included dataset setup, preprocessing, resizing images to 32×32 and 64×64 resolutions, and conducting baseline experiments using Logistic Regression, XGBoost, and Random Forest. He also contributed to the FFT-based experiment, SVM scaling and wider grid-search experiments using pipeline-based tuning, and CNN fine-tuning in the intermediate modeling stage. In addition, he supported workflow documentation, organization of experimental results, formatting of tables and references, and preparation of the final IEEE-style report.

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